

# Aditya Singhal

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## SUMMARY

Software Engineer with experience building backend platforms, full-stack applications, AI-enabled services, and simulation-backed engineering systems. Strong with Python, FastAPI, JavaScript, React, MongoDB, PostgreSQL, Docker, AWS, and system architecture, with additional exposure to embedded C and autonomous-system simulation workflows.

## SKILLS

**Languages::** Python, JavaScript, TypeScript, SQL, HTML/CSS, Embedded C, Java

**Frontend::** React, Vite, Bun, npm, tailwind

**Backend::** FastAPI, REST APIs, MongoDB, PostgreSQL, Redis, NocoDB, RabbitMQ, Express, Node

**DevOps/Infra::** Docker, AWS, Linux, Git, Server Architecture, Scaling

**Simulations::** SUMO, CARLA, ROS, Gazebo

## EXPERIENCE

### ITEL Foundation, IIT Madras Research Park

Chennai, India

*Project Associate*

*May 2025 – Present*

- Designed software and architecture workflows for an autonomous Personal Rapid Transit (PRT) system spanning planning logic, simulation-backed validation, and system integration.
- Led design discussions around backend architecture and server scaling while collaborating with multidisciplinary engineering teams.
- Guide team of 4 and trained new joiners in modern industry standard practices
- Built SaaS edtech RAG solutions for clients, designing information retrieval workflows, delivery coordination, and technical requirement translation.

### Tagglabs Experiential

Gurugram, India

*AI Solutions Engineer*

*Jun 2023 – Feb 2025*

- Developed backend services and product infrastructure using FastAPI, Docker, AWS, Redis, and MongoDB for AI and production systems.
- Worked across AI, web, and hardware-integrated projects; improved system latency by 40% through service and workflow refinements.
- Built internal automation workflows with n8n and NocoDB; contributed to scalable product delivery and operational efficiency.
- Led a team of 6 engineers and delivered 100+ AI, web, and hardware-integrated projects within tight timelines.
- Built automated hiring pipeline using N8N and NocoDB processing 750+ applications and reducing hiring time by 85%.

### e-Yantra, IIT Bombay

Mumbai, India

*eYSIP Intern*

*Jun 2022 – Jul 2022*

- Built robotics software workflows for obstacle avoidance and swarm control using ROS and Gazebo in a simulation-heavy environment.

## PROJECTS

- Full-Stack Face Analysis App:** a containerized end-to-end application with React frontend and FastAPI backend, integrating a custom trained ViT model that analyzes facial images to detect skin conditions with 92% accuracy and recommends personalized skincare products based on identified ailments.
- Forecasting Pipeline:** A full MLOps pipeline featuring multiple Prophet models for stock price prediction. Implemented MANGO optimization library, reducing training time by 35% while improving forecast accuracy by 22%.

## EDUCATION & CERTIFICATIONS

### Amrita University

Coimbatore, Tamil Nadu, India

*Bachelor of Technology - Computer Science and Engineering (AI):*

*Jul 2019 - Aug 2023*

**Relevant Coursework:** Operating Systems, Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Deep Learning, Data Visualization, Robotics, Statistical Analysis, Database Systems

## RESEARCH WORK

### KAP of Rabies Post Exposure Prophylaxis among Young Doctors

*Biostatistician*

*J Med Soc 2023*

- Performed statistical analysis, hypothesis testing, and data visualization for medical research on rabies post-exposure prophylaxis, contributing to published journal research.

### Quiz in PG Teaching: Are we adapted to new methods

*Biostatistician*

*J Med Soc 2023*

- Conducted statistical modeling and data analysis to evaluate postgraduate medical teaching methods, contributing to measurable improvements in educational outcomes.